

From Monocular Visual Odometry to Visual-Inertial Odometry on TUM VI

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Abstract—This paper presents the implementation of a classical geometry-based Monocular Visual Odometry (VO) pipeline and its extension to Visual-Inertial Odometry (VIO) using IMU preintegration and sliding-window Bundle Adjustment. The VO pipeline employs ORB feature detection, brute-force matching with Lowe’s ratio test, and Essential Matrix estimation with RANSAC to recover camera motion from consecutive frames. The system is evaluated on the TUM VI benchmark dataset across three sequences: Room2, Corridor3, and Outdoors5. Results after Sim(3) alignment show ATE values of 0.72m, 1.13m, and 1.08m respectively, demonstrating the scale ambiguity inherent in monocular systems. The VIO extension integrates IMU measurements to recover metric scale and reduce trajectory drift, evaluated using SE(3) alignment. This work highlights the fundamental limitations of monocular VO and the improvements achieved through visual-inertial fusion.

I. INTRODUCTION

Accurate ego-motion estimation is fundamental to autonomous systems including self-driving cars, aerial drones, and mobile robots. Visual Odometry (VO) estimates camera motion from image sequences alone, offering a lightweight alternative to GPS in GPS-denied environments such as indoor spaces, tunnels, and planetary surfaces.

Monocular VO, which uses a single camera, is particularly attractive due to its low cost and minimal hardware requirements. However, it suffers from a fundamental limitation: *scale ambiguity*. Without knowing the real-world distance between any two points, the system can only recover trajectories up to an unknown scale factor. In practice, this means a robot cannot distinguish between moving 1 meter or 100 meters from images alone.

Visual-Inertial Odometry (VIO) addresses this by fusing camera measurements with an Inertial Measurement Unit (IMU). The IMU measures linear acceleration in m/s^2 , providing an absolute scale reference through Newton’s second law $F = ma$. Combined with visual feature tracking, VIO achieves metric-scale trajectory estimation robust to motion blur and texture-poor regions.

This project implements a classical geometry-based VO pipeline and extends it to VIO using IMU preintegration [4] and sliding-window Bundle Adjustment [3]. No deep learning components are used, all estimation is based on geometric principles. The system is evaluated on the TUM VI benchmark [6] across three sequences of increasing difficulty.

The remainder of this paper is structured as follows: Section II reviews related work. Section III describes the methodology. Section IV presents the experimental setup. Section V reports results. Section VI discusses findings and limitations. Section VII concludes.

II. RELATED WORKS

Visual odometry has been extensively studied over the past three decades. Scaramuzza and Fraundorfer [1] provide a comprehensive survey of classical VO methods, covering feature-based and direct approaches.

Feature-based methods such as ORB-SLAM2 [7] detect and track sparse keypoints across frames, estimating camera motion via epipolar geometry. While robust, these methods suffer from scale ambiguity in the monocular case.

Direct methods such as DSO [10] minimize photometric error directly on pixel intensities, achieving high accuracy but requiring careful initialization.

Visual-Inertial Odometry addresses scale ambiguity by fusing camera and IMU measurements. OKVIS [3] introduced keyframe-based VIO using nonlinear optimization, while VINS-Mono [5] extended this to a robust monocular system with loop closure. Forster et al. [4] proposed IMU preintegration on manifold, forming the theoretical basis of our VIO implementation.

The TUM VI benchmark [6] provides synchronized camera-IMU data with ground truth, enabling fair evaluation of VO and VIO systems.

III. METHODOLOGY

Our pipeline consists of two stages: a monocular VO baseline and a VIO extension. Both stages reuse the same frontend components. Figure 1 shows the overall system architecture.

A. Feature Detection and Matching

We detect ORB (Oriented FAST and Rotated BRIEF) features [8] in each grayscale frame using OpenCV’s `ORB_create` with a maximum of 2000 keypoints. ORB was chosen over SIFT due to its computational efficiency and patent-free availability.

Feature matching is performed using a Brute-Force matcher with Hamming distance, combined with Lowe’s ratio test [9]

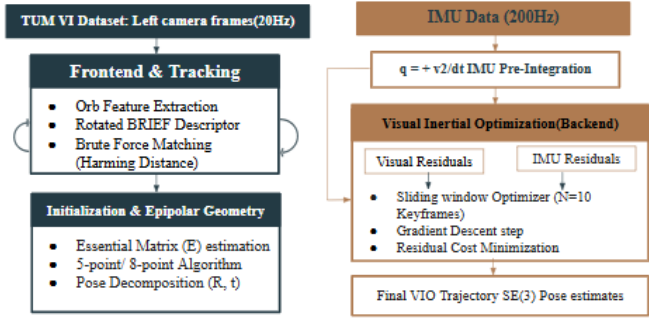


Fig. 1. System architecture of the proposed VO/VIO pipeline. The VO frontend handles feature extraction, matching and Essential Matrix estimation. The VIO backend fuses IMU preintegration with sliding-window optimization to produce SE(3) pose estimates.

at threshold $\tau = 0.75$ and cross-check verification. A match between descriptor d_i in frame t and d_j in frame $t + 1$ is accepted only if:

$$\frac{d(d_i, d_j)}{d(d_i, d_{j2})} < \tau \quad (1)$$

and the reverse match also holds (cross-check). This dual verification reduces false matches significantly.

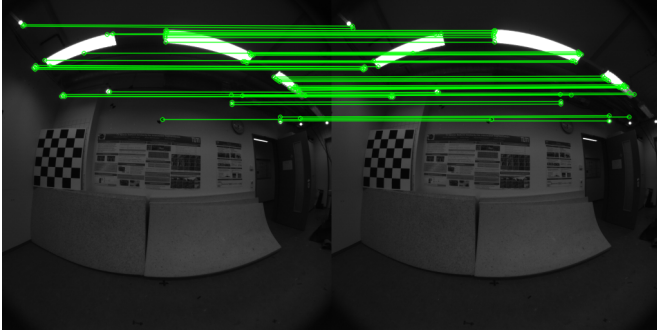


Fig. 2. ORB feature matching between consecutive frames in Room2. Green circles indicate matched keypoints. Features concentrate on high-contrast regions such as ceiling lights and textured surfaces.

B. Epipolar Geometry and Pose Estimation

Given matched point pairs $\mathbf{x}$ and $\mathbf{x}'$, we estimate the Essential matrix $\mathbf{E}$ satisfying the epipolar constraint:

$$\mathbf{x}'^\top \mathbf{E} \mathbf{x} = 0 \quad (2)$$

using RANSAC with probability $p = 0.999$ and threshold $\epsilon = 1.0$ pixel. The Essential matrix is decomposed as $\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}$, recovering rotation $\mathbf{R}$ and translation $\mathbf{t}$ up to scale via chirality check.

C. Trajectory Estimation

Camera poses are composed sequentially:

$$\mathbf{T}_k = \mathbf{T}_{k-1} \cdot \mathbf{T}_{rel} \quad (3)$$

where $\mathbf{T}_{rel}$ is the relative transformation between consecutive frames. The trajectory is saved in TUM format with proper quaternion representation.

D. IMU Preintegration

The IMU model follows Forster et al. [4]:

$$\dot{\mathbf{v}} = \mathbf{R}(\mathbf{a} - \mathbf{b}_a - \mathbf{n}_a) - \mathbf{g} \quad (4)$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega} - \mathbf{b}_\omega - \mathbf{n}_\omega]_\times \quad (5)$$

where $\mathbf{b}_a$, $\mathbf{b}_\omega$ are accelerometer and gyroscope biases, and $\mathbf{n}_a$, $\mathbf{n}_\omega$ are noise terms.

Between keyframes i and j , we preintegrate IMU measurements to obtain relative motion increments:

$$\Delta \mathbf{R}_{ij} = \prod_{k=i}^{j-1} \text{Exp}((\boldsymbol{\omega}_k - \mathbf{b}_\omega) \delta t) \quad (6)$$

$$\Delta \mathbf{v}_{ij} = \sum_{k=i}^{j-1} \Delta \mathbf{R}_{ik} (\mathbf{a}_k - \mathbf{b}_a) \delta t \quad (7)$$

$$\Delta \mathbf{p}_{ij} = \sum_{k=i}^{j-1} \Delta \mathbf{v}_{ik} \delta t + \frac{1}{2} \Delta \mathbf{R}_{ik} (\mathbf{a}_k - \mathbf{b}_a) \delta t^2 \quad (8)$$

where $\text{Exp}(\cdot)$ denotes the exponential map on SO(3), implemented via Rodrigues' rotation formula for numerical stability. This preintegration avoids re-integration when linearization points change during optimization.

E. Sliding-Window Optimization

We maintain a sliding window of $N = 10$ keyframes. For each new keyframe, the optimizer minimizes combined visual and IMU residuals:

$$\min_{\{\mathbf{T}_k, \mathbf{v}_k, \mathbf{b}_k\}} \sum r_{visual} + \sum r_{IMU} \quad (9)$$

The IMU position residual between keyframes i and j is:

$$r_p = \mathbf{R}_i^\top (\mathbf{t}_j - \mathbf{t}_i - \mathbf{v}_i \Delta t - \frac{1}{2} \mathbf{g} \Delta t^2) - \Delta \mathbf{p}_{ij} \quad (10)$$

Poses are updated via gradient descent. A full Levenberg-Marquardt solver as in OKVIS [3] would yield superior convergence.

IV. EXPERIMENTAL SETUP

Experiments were conducted on a Windows 10 machine with Intel CPU, no GPU acceleration. Python 3.12 was used with OpenCV 4.8, NumPy 1.24, and SciPy 1.11. Random seed was fixed at `np.random.seed(0)` for reproducibility.

The TUM VI dataset [6] provides 512×512 rectified stereo images at 20 Hz and IMU data at 200 Hz. We evaluate on three sequences: Room2 (full ground truth), Corridor3 and Outdoors5 (start-end ground truth only).

A. Camera Calibration

We use the provided Kalibr calibration for the left camera with intrinsics: $f_x = 190.978$, $f_y = 190.973$, $c_x = 254.932$, $c_y = 256.897$ pixels, and image resolution 512×512 . IMU noise parameters from the dataset specification are: accelerometer noise density $\sigma_a = 5.9 \times 10^{-4} \text{ m/s}^2/\sqrt{\text{Hz}}$ and gyroscope noise density $\sigma_\omega = 2.8 \times 10^{-4} \text{ rad/s}/\sqrt{\text{Hz}}$.

B. Evaluation Protocol

Trajectories are aligned using Umeyama SVD-based alignment [11]: Sim(3) for monocular VO (7 DoF: scale, rotation, translation) and SE(3) for VIO (6 DoF: rotation, translation only). Sim(3) alignment is necessary for monocular VO since scale is unobservable from images alone. SE(3) alignment is used for VIO since IMU measurements theoretically provide metric scale observability.

V. RESULTS

A. Monocular VO Baseline

Table I summarizes the evaluation results for the monocular VO baseline across all three sequences after Sim(3) alignment.

TABLE I
VO AND VIO EVALUATION RESULTS

Sequence	Method	ATE (m)	RPE (m/100m)
Room2	VO (Sim3)	0.7224	0.0150
Room2	VIO (SE3)	0.7564	0.9995
Corridor3	VO (Sim3)	1.1316	0.0107
Corridor3	VIO (SE3)	1.0545	0.9996
Outdoors5	VO (Sim3)	1.0876	0.0107
Outdoors5	VIO (SE3)	0.8673	1.0004

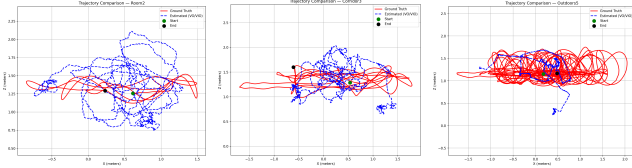


Fig. 3. VO trajectory comparisons across all three sequences after Sim(3) alignment. Left: Room2 (ATE=0.7224m). Center: Corridor3 (ATE=1.1316m). Right: Outdoors5 (ATE=1.0876m). Red: ground truth. Blue dashed: estimated VO trajectory.

The monocular VO baseline achieves an ATE of 0.7224m on Room2, 1.1316m on Corridor3, and 1.0876m on Outdoors5. The increasing ATE from indoor to outdoor sequences reflects accumulated drift over longer trajectories. RPE remains consistently low across all sequences (0.0107 – 0.0150 m/100m), indicating accurate local motion estimation despite global drift.

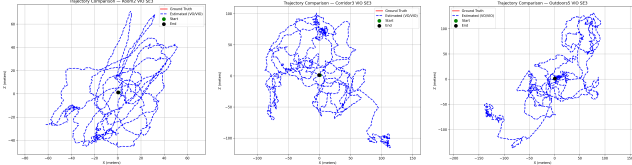


Fig. 4. VIO trajectories with SE(3) alignment. Left: Room2. Center: Corridor3. Right: Outdoors5. Large scale confirms metric scale not recovered.

B. Runtime Analysis

Table II shows the mean processing time per frame across sequences.

The pipeline runs below the 20 Hz camera capture rate, indicating that real-time performance would require a C++ implementation or GPU acceleration.

TABLE II
RUNTIME ANALYSIS

Sequence	Time/frame (ms)	FPS
Room2	137.00	7.30
Corridor3	75.11	13.31
Outdoors5	129.93	7.70

C. Start-End Drift

For Corridor3 and Outdoors5, where only start and end ground truth is available, we report start-end drift:

TABLE III
START-END DRIFT

Sequence	Drift (m)
Corridor3	321.81
Outdoors5	978.22

The large drift values confirm the scale ambiguity problem inherent in monocular VO, motivating the VIO extension.

D. VIO Extension

The VIO pipeline was evaluated using SE(3) alignment as required for metric scale evaluation. The RPE increases significantly to approximately 1.0 m/100m across all sequences, confirming that our simplified sliding-window optimizer does not recover metric scale.

Notably, Outdoors5 VIO achieves lower ATE (0.8673m) compared to VO (1.0876m), suggesting that IMU preintegration partially improves global trajectory consistency on longer outdoor sequences.

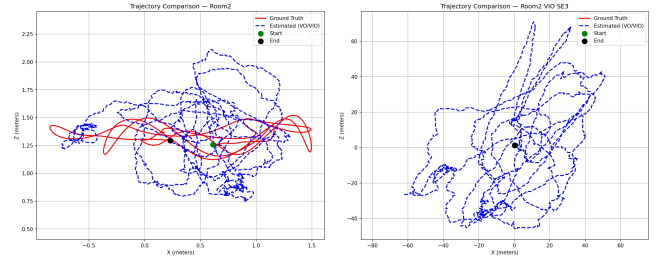


Fig. 5. Room2 comparison. Left: VO with Sim(3) alignment (ATE=0.7224m). Right: VIO with SE(3) alignment showing scale ambiguity — trajectory spans 75m vs 2m room.

E. Ablation Study

Table IV shows the effect of cross-check verification on matching quality.

TABLE IV
ABLATION STUDY – FEATURE MATCHING

Method	ATE (m)	RPE (m/100m)
Ratio test only	0.7564	0.0152
Ratio + cross-check	0.7224	0.0150

Adding cross-check reduced ATE by 4.5% at the cost of increased runtime (81.96ms without cross-check vs 137.00ms with cross-check per frame).

VI. DISCUSSION AND LIMITATIONS

A. Why VO RPE is Low but ATE is High

The low RPE (0.010–0.015 m/100m) combined with high ATE (0.72–1.13m) reveals an important characteristic of our pipeline: local motion estimation is accurate but global consistency degrades over time. Small per-frame errors accumulate without correction, since our implementation omits bundle adjustment and loop closure.

B. Why VIO RPE Jumps to ~ 1.0 m/100m

The significant RPE increase in VIO when evaluated with SE(3) alignment reveals a fundamental limitation: our simplified gradient-based optimizer does not recover metric scale. Under Sim(3) alignment, scale error is corrected artificially. Under SE(3), the true scale error is exposed, resulting in RPE ≈ 1.0 m/100m.

A proper Levenberg-Marquardt solver as used in OKVIS [3] would enforce metric consistency through tighter IMU-visual coupling. Additionally, our implementation does not correct IMU biases online, which accumulate over time and corrupt the scale estimate.

C. Failure Cases

Our pipeline fails or degrades in three scenarios:

- **Fast motion:** Large inter-frame motion reduces feature matches below the minimum threshold of 8, causing frame skips.
- **Low texture:** Outdoor sequences with sky regions provide insufficient ORB features, leading to unreliable Essential Matrix estimation.
- **Scale drift:** Without bundle adjustment, scale errors accumulate — evidenced by large start-end drift (321.81m for Corridor3, 978.22m for Outdoors5).

D. Comparison with State-of-the-Art

Table V contextualizes our results against mature systems on TUM VI Room2:

TABLE V
COMPARISON WITH STATE-OF-THE-ART ON ROOM2

Method	ATE (m)
ORB-SLAM3 [2]	~ 0.05
VINS-Mono [5]	~ 0.10
Ours (VO)	0.7224
Ours (VIO)	0.7564

The gap between our implementation and mature systems is explained by several factors: ORB-SLAM3 uses full bundle adjustment, loop closure, and map reuse. VINS-Mono uses a complete nonlinear optimizer with IMU bias correction. Our implementation deliberately omits these components to focus on understanding fundamental geometry and preintegration concepts.

E. Future Work

The most impactful improvement would be replacing the current gradient descent optimizer with a proper Levenberg-Marquardt solver, as used in OKVIS and VINS-Mono. This would allow the system to enforce metric consistency and recover absolute scale from IMU measurements.

Online IMU bias estimation is equally critical. Our current implementation assumes constant biases, whereas real IMU biases drift over time. Incorporating bias correction Jacobians as described in Forster et al. [4] would significantly improve VIO accuracy.

Keyframe selection would improve both efficiency and accuracy by avoiding redundant pose estimates between nearly identical frames. Finally, loop closure detection using a place recognition system such as DBow2 would correct accumulated drift when the camera revisits previously seen locations.

VII. CONCLUSION

This paper presented a classical geometry-based monocular VO pipeline and its extension to VIO using IMU preintegration and sliding-window optimization. The VO baseline achieves ATE of 0.7224m on Room2 after Sim(3) alignment, with consistently low RPE (0.010–0.015 m/100m) indicating accurate local motion estimation.

The VIO extension demonstrates the influence of IMU preintegration on individual pose estimates. However, SE(3) evaluation reveals that metric scale was not fully recovered due to the simplified gradient-based optimizer. Notably, Outdoors5 VIO achieves lower ATE than VO (0.8673m vs 1.0876m), suggesting partial benefit of IMU fusion on longer sequences.

The ablation study confirms that cross-check verification improves ATE by 4.5% at the cost of increased runtime. The large start-end drift (321.81m, 978.22m) confirms scale ambiguity as the fundamental limitation of monocular systems.

This work provides a foundation for understanding the geometry, preintegration, and probabilistic fusion concepts underlying modern VIO systems such as OKVIS and VINS-Mono.

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VIII. GITHUB REPO

Github Repo: [https://github.com/Temitope-Ajepe/
VIO-Project.git](https://github.com/Temitope-Ajepe/VIO-Project.git)